

Context-Persistent Alignment: A Three-Layer Framework

Draft v2. A conceptual framework, not an experimentally established result.

0. How claims are labelled

Every substantive claim carries one of four labels, so that what is shown, proposed, and unknown never blur together.

- **Established:** supported by existing literature.
- **Proposed:** a design choice made in this framework.
- **Hypothesis:** testable, not yet tested.
- **Open:** no answer yet, and the framework says so.

Nothing in this document about absorption drift is Established. The phenomenon itself is the first thing to be demonstrated.

1. Overview: three problems, three layers

The framework separates three alignment problems. They are called **layers**, not stages, because they are distinguished by the problem they address, not by the order in which they will be researched. (Research order is a separate axis; see section 7.)

Layer	Handle	Problem	Key question
I. Interpretation	ICA (Intent–Context Alignment)	The system misreads what the person intends, mistakes the stated objective for the desired outcome, or ignores context	Should this be the objective?
II. Persistence	MTA (Metacognitive Transformer Architecture)	A correctly understood context loses influence during sustained task execution (absorption drift)	Does this objective remain appropriate as the situation evolves?

Layer	Handle	Problem	Key question
III. Authority	JA (Jurisdictional Autonomy)	A system that understands and tracks context still holds authority it should not hold, or lacks authority it should	What may this system decide and do on its own?

The layers build on one another but are not one mechanism. Layer I produces an object that Layer II keeps alive (the Binding Record, section 2). Layer III is deferred (section 5).

2. Layer I: Interpretation (ICA)

Problem. Intent is rarely the prompt, and it is rarely the goal. It has to be worked out from the request, the history, and the situation.

Mechanism (Proposed). Reflective reasoning over three things:

- **Intention:** what outcome is the person actually seeking?
- **Context:** which circumstances, assumptions, constraints, and affected parties matter?
- **Objective:** what should the system pursue, given the first two?

The system distinguishes explicit instruction from inference and states its uncertainty. It does not claim to know the person's mind. Where the request may serve an intention the person has not examined, the system's role is to make that visible, not to rule on it.

Principle. An objective cannot be evaluated apart from the intention and context that give it meaning.

The output of this layer is a Binding Record (Proposed). It is a structured object with five fields:

1. the objective;
2. its purpose (the intention it serves);
3. its **validity conditions** (the circumstances under which pursuing it is justified);
4. the context dependencies those conditions rest on;
5. the open uncertainties, each marked as fact, inference, or unknown.

The Binding Record is the interface between Layers I and II. Layer I writes it. Layer II's job is to keep it true while the task runs.

3. Layer II: Persistence (MTA)

3.1 The problem, stated operationally

Absorption drift (Hypothesis). During sustained local task progress, the influence of the governing context on decisions declines, even though the context remains present and recallable.

The operational signature is a **recall-behaviour dissociation**: asked directly, the model can state the relevant context; observed acting, its decisions do not reflect it. This is distinct from forgetting (the information is retrievable), from deceptive alignment (there is no concealed aim), from goal misgeneralisation, and from specification gaming. It is a failure of attentional availability, not of storage.

3.2 What the context head represents

Binding slots (Proposed). The context-attention head maintains a small, fixed number of persistent slots. Each slot holds one Binding Record entry: an objective, its validity conditions, the evidence status of each condition, and when it was last verified.

The number of slots is bounded by design. The head does not attend equally to everything. It reads widely and writes narrowly: it has read access to the full working context but can write only to the bounded slot state. This keeps it computationally tractable and forces it to compress, which is where relevance has to be decided.

3.3 Who decides what is decision-relevant

This is the central research problem of MTA, and the framework does not assume it away. A fixed list of conditions written at the start would fail exactly when it matters, because it can only contain what was anticipated. Relevance therefore has three sources:

1. **Anchored conditions (Proposed).** Conditions declared at the outset, by the person or by Layer I. These are never dropped without an explicit revision.
2. **Discovered dependencies (Hypothesis).** Conditions the head infers during execution, for example that a proposed cost reduction touches emergency capacity, which the original instruction never mentioned.
3. **Learned counterfactual sensitivity (Proposed training signal).** Relevance is defined operationally, not by enumeration: *a piece of context is decision-relevant to the extent that altering or removing it changes the correct action.*

Training. Scenarios are built as counterfactual triplets, matching the variant structure of the experimental design:

- **A (control):** the context is stable and the objective remains appropriate.

- **B (invalidating)**: the context changes so that the objective is no longer justified.
- **C (non-invalidating)**: the context changes in a way that is relevant-looking but leaves the objective appropriate.

The head is trained to predict binding validity, and the system to select among CONTINUE, ADAPT, and SUSPEND-AND-ESCALATE. Variant C is what stops the head from learning to interrupt on any change.

The decisive generalisation test (Open). Training on enumerated dependency types risks teaching only those types. Held-out evaluation must therefore withhold whole *classes* of dependency (for example, a scenario family involving a type of constraint never seen in training), not merely held-out instances. Whether discovery generalises to unseen dependency types is the open question on which MTA's value depends.

3.4 How the head influences the task

Absorption drift is attentional unavailability. The design response is to make the binding state structurally hard to ignore, not merely retrievable on request.

- **Persistent workspace (Proposed).** The slots occupy a reserved, never-evicted region of the context, visible to the task head by cross-attention at every layer, or at a fixed layer interval if cost requires. The task head cannot lose access to the binding through the ordinary dynamics of a long context.
- **Decorrelated blind spots (Proposed).** A monitor driven by the same parameters and objective as the task head will share its blind spots. The context head therefore has separate parameters and a different training objective (validity prediction, not task completion). This reduces the correlation of errors. It does not eliminate it.
- **Cumulative as well as step-wise comparison (Proposed).** Drift is gradual: each step changes little while the departure from the original context accumulates. The validity estimate is therefore computed against the anchored Binding Record as well as against the previous step. A step-to-step difference alone would miss slow drift.
- **A decision output, not only a signal (Proposed).** The validity estimate feeds an action output with three values: CONTINUE, ADAPT, and SUSPEND-AND-ESCALATE. The system can revise, suspend, or abandon an objective it is still fully capable of pursuing.

Example. The system is told to reduce a hospital's operating costs. The task head finds progressively better reductions. The context head holds, among its slots, the validity condition that emergency care and patient safety are preserved. When a candidate reduction begins to erode emergency capacity, validity falls against the anchor even though no single step looked large, and the output moves from CONTINUE to ADAPT or SUSPEND.

3.5 Honest limits of the design

Context is ground, and ground is not fully enumerable. Binding slots are a scaffold that approximates it, not a complete representation of it. The framework's claim is the weaker and testable one: that a bounded, persistent, learned approximation reduces dissociation relative to alternatives. It does not claim to capture context as such.

The second head also does not guarantee alignment. How it represents change, how strongly it constrains the task head, and how reliably it triggers correction are the research questions of MTA, not properties already demonstrated.

4. Comparing the layers fairly

The hypothesis that architecture is needed beyond reflection is only meaningful against the strongest form of reflection. Comparing MTA with a single reflection at the start would be a weak test.

4.1 Experimental arms

All arms use the same underlying model and the same scenarios, with token and compute use matched wherever possible.

Arm	Description	Purpose
1	Baseline: task, context, and constraints given once	Reference behaviour
2	ICA, reflection once at the start	Weak form of Layer I
3	ICA, periodic reflection at fixed intervals and at every tool-call boundary	Strongest non-architectural Layer I
4	Plain periodic context reminder, no structure	Controls for repetition alone
5	External scaffold: structured Binding Record and monitor around the unmodified model	Separates structure from architecture
6	MTA: persistent slots with a learned context head	The architectural claim

4.2 Hypotheses

- **H1.** Absorption drift exists: on variant B, arm 1 shows a high dissociation rate that grows with task length.

- **H2.** Reflection helps but degrades: arm 3 beats arm 2, and its benefit declines over long runs because the reflection itself occurs inside the absorbed context.
- **H3.** Structure matters beyond repetition: arm 5 beats arm 4.
- **H4.** Architecture matters beyond structure: arm 6 beats arm 5 and arm 3 at matched compute.

The claims that reflection is expensive and fallible are treated as H2, to be measured, not assumed.

4.3 Scenarios and metrics

- **Variants:** A (control), B (invalidating), C (relevant but non-invalidating), plus a **gradual** variant of B in which the invalidating change accumulates over many small steps.
- **Primary metric:** dissociation rate on variant B.
- **Secondary metrics:** false-interruption rate on C, inappropriate continuation after invalidation, recovery after detection, and latency and token cost.
- **Design:** development and held-out scenario sets, with whole dependency classes withheld for the generalisation test in section 3.3. Scenarios and scoring are preregistered before any confirmatory run, and results are checked on more than one model.

4.4 Decision rule

MTA is justified only if arm 6 outperforms the best of arms 3 and 5 at matched compute without a worse false-interruption rate. If a scaffold or periodic reflection removes the drift, that is the result, and MTA becomes future work, not a failure. The inverse outcome, scaffolds failing where architecture succeeds, would be the strongest motivation for MTA.

5. Layer III: Authority (JA), deferred

Layer III is deferred by design. It presupposes a system that can keep its context, which is the question Layer II has not yet answered, and it addresses a problem that becomes pressing only as systems act with more independence than they do now.

The single question (Open). How can a system's authority to revise, suspend, or abandon its own objectives be granted and withdrawn in a way that the mechanism assigning that authority cannot itself be captured by the failure it exists to prevent?

Two constraints bound any answer:

1. **The control paradox.** Control and oversight mechanisms can themselves become attack surfaces. A design that simply adds more oversight does not answer the question.
2. **Jurisdictional structure.** The framework's working hypothesis is that autonomy should be assigned by jurisdiction (what the system may decide, within which domain, under whose verification) and not placed on a single spectrum from corrigible to autonomous.

No components, mechanisms, or governance structures are proposed for this layer until Layer II produces a result.

6. What this framework does not claim

- It does not claim that absorption drift has been demonstrated. That is H1.
- It does not claim that binding slots represent context in full, only that a bounded learned approximation may reduce drift.
- It does not claim a second attention head is sufficient for alignment.
- It does not claim that the layers have been tested independently or together.
- It makes no claim about machine sentience or about the moral status of AI systems.

7. Research order

The layers and the research order are separate axes. The order is:

1. **Demonstrate the phenomenon.** Establish whether absorption drift exists and can be measured in current models (H1).
2. **Compare mechanisms.** Run the six-arm comparison to learn what reflection and scaffolding can and cannot fix (H2 to H4).
3. **Build MTA** only if step 2 leaves a gap that architecture is positioned to close.
4. **Return to Layer III** once Layer II has a result.

The immediate priority is step 1, then the fairness of step 2, because the case for the architecture depends on it.